

TITLE AND DATE

Document name: Project#4 UserRTOS Lab Report

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LEAD ENGINEER:

Nathan Demory, UMBC

STAKEHOLDERS:

Prof Kidd, Instructor, University of Maryland Baltimore County, Baltimore, Maryland, USA

MD Safwan Zaman, TA, University of Maryland Baltimore County, Baltimore, Maryland, USA

HIGH-LEVEL DESCRIPTION:

This is the project document for project three of CMPE 311 class spring 2026. It extends project two (PROJECT-ADDMOTOR) by replacing the cyclic executive round-robin scheduler with a FreeRTOS-based implementation. The hardware design and customer requirements are unchanged. The software has been refactored into four FreeRTOS tasks: Task_Serial, Task_LED1, Task_LED2, and Task_Fan. This document records the updated high-level requirement, new testing requirements, and confirms all previous tests remain satisfied.

DESCRIPTION:

This project replaces the cyclic executive scheduler from PROJECT-ADDMOTOR with FreeRTOS running on the Arduino Uno R3. All previous functionality is preserved: serial-controlled LED blinking and button-controlled fan speed. Each function is now implemented as a separate FreeRTOS task with appropriate priorities. A mutex protects shared serial state between tasks. The hardware circuit (MOSFET, diode, capacitor, button, motor) is identical to the previous project.

REFERENCES:

- 311projectFreeRTOS.pdf – The definition of the project this document addresses
- ProjectAddMotor – The previous project upon which this project is based
- Demory.cmpe311.spring26.project#2 – Previous project report
- Arduino_FreeRTOS Library – https://github.com/feilipu/Arduino_FreeRTOS_Library
- Arduino UNO R3 Product Reference Manual SKU A000066, 12/03/2024

DEFINITIONS AND ABBREVIATIONS:

See Demory.cmpe311.spring26.project#2 for previously defined terms.

"RTOS" – Real-Time Operating System; an OS designed to process data in real time with deterministic scheduling.

"FreeRTOS" – A widely-used open-source RTOS kernel ported to AVR via the Arduino_FreeRTOS library.

"Task" – A FreeRTOS thread of execution, replacing the functions in the previous cyclic executive.

"Mutex" – Mutual exclusion semaphore used to protect shared resources between tasks.

"vTaskDelay" – FreeRTOS API call to block a task for a given number of ticks, yielding CPU to other tasks.

"Tick" – The FreeRTOS time unit; on Arduino Uno with default settings approximately 15ms per tick.

CONVENTIONS:

See Demory.cmpe311.spring26.project#2 — conventions are unchanged.

CUSTOMER REQUIREMENTS:

C1–C12: Unchanged. See Demory.cmpe311.spring26.project#2.

HIGH-LEVEL TECHNICAL REQUIREMENTS:

HL.1–HL.18: Unchanged. See Demory.cmpe311.spring26.project#2.

HL.19. The system and application software will be built upon a RTOS (Real-time Operating System). (added – Kidd, 4/1/2026 / implemented – Demory, 4/23/2026)

EDUCATIONAL REQUIREMENTS:

E.1–E.4: Unchanged. See Demory.cmpe311.spring26.project#2.

E.5. Verify that all previous tests still work for the current project. (added – Kidd, 4/1/2026)

TESTING AND VALIDATION REQUIREMENTS:

T.1–T.12: Unchanged. See Demory.cmpe311.spring26.project#2. All previous tests verified and still satisfied — see video.

T.13 – System runs on FreeRTOS: confirm four tasks created (Task_Serial, Task_LED1, Task_LED2, Task_Fan). (added – Demory, 4/23/2026)

T.14 – Serial LED dialogue functions correctly under FreeRTOS with no degradation from previous behavior. (added – Demory, 4/23/2026)

T.15 – LED blinking is independent and correct under FreeRTOS scheduling. (added – Demory, 4/23/2026)

T.16 – Fan button cycling functions correctly under FreeRTOS with no degradation from previous behavior. (added – Demory, 4/23/2026)

T.17 – All previous tests T.1–T.12 pass without modification under FreeRTOS implementation. (added – Demory, 4/23/2026)

DESIGN:

DESIGN PRE-REQUISITES:

- ELEGOO Arduino Uno R3 clone
- Arduino IDE 2.3.3 or better
- Arduino_FreeRTOS library installed in Arduino IDE
- Breadboard
- Two LEDs
- DC motor / fan
- N-channel MOSFET (e.g. IRF540N)
- 1N4007 flyback diode
- 220Ω resistor (gate), 10kΩ resistor (pull-down)
- 100μF capacitor (smoothing)
- 9V battery + connector
- Momentary pushbutton
- Connective wires

DESIGN OVERVIEW:

The cyclic executive scheduler (taskList[] round-robin) from project#2 has been replaced with four FreeRTOS tasks. The hardware circuit is identical to project#2.

Task priorities:

- Task_Serial — priority 2 (highest): must not miss serial input
- Task_LED1, Task_LED2, Task_Fan — priority 1: equal, preemptable by Task_Serial

LED tasks use `vTaskDelay(pdMS_TO_TICKS(interval/2))` for accurate, non-blocking blink timing. The fan task uses `vTaskDelay` for debounce. A mutex (`xSerialMutex`) protects shared variables `interval1`, `interval2`, `selectedLED`, and `waitingForInterval`.

DEVELOPMENT PLATFORM:

- Uses Arduino IDE 2.3.3 or later
- Codes in C++ with Arduino_FreeRTOS library

Code is attached in submission and located on GitHub repository.

Video is attached to submission.

TESTING:

Test Performed	Results
T1	Satisfied – see video (verified on FreeRTOS)
T2	Satisfied – see video (verified on FreeRTOS)
T3	Satisfied – see video (verified on FreeRTOS)
T4	Satisfied – see video (verified on FreeRTOS)
T5	Satisfied – see video (verified on FreeRTOS)
T6	Satisfied – see video (verified on FreeRTOS)
T7	Satisfied – see video (verified on FreeRTOS)
T8	Satisfied – see video (verified on FreeRTOS)
T9	Satisfied – see video (verified on FreeRTOS)
T10	Satisfied – see video (verified on FreeRTOS)
T11	Satisfied – see video (verified on FreeRTOS)
T12	Satisfied – see video (verified on FreeRTOS)
T13	Satisfied – see video
T14	Satisfied – see video
T15	Satisfied – see video
T16	Satisfied – see video
T17	Satisfied – see video

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